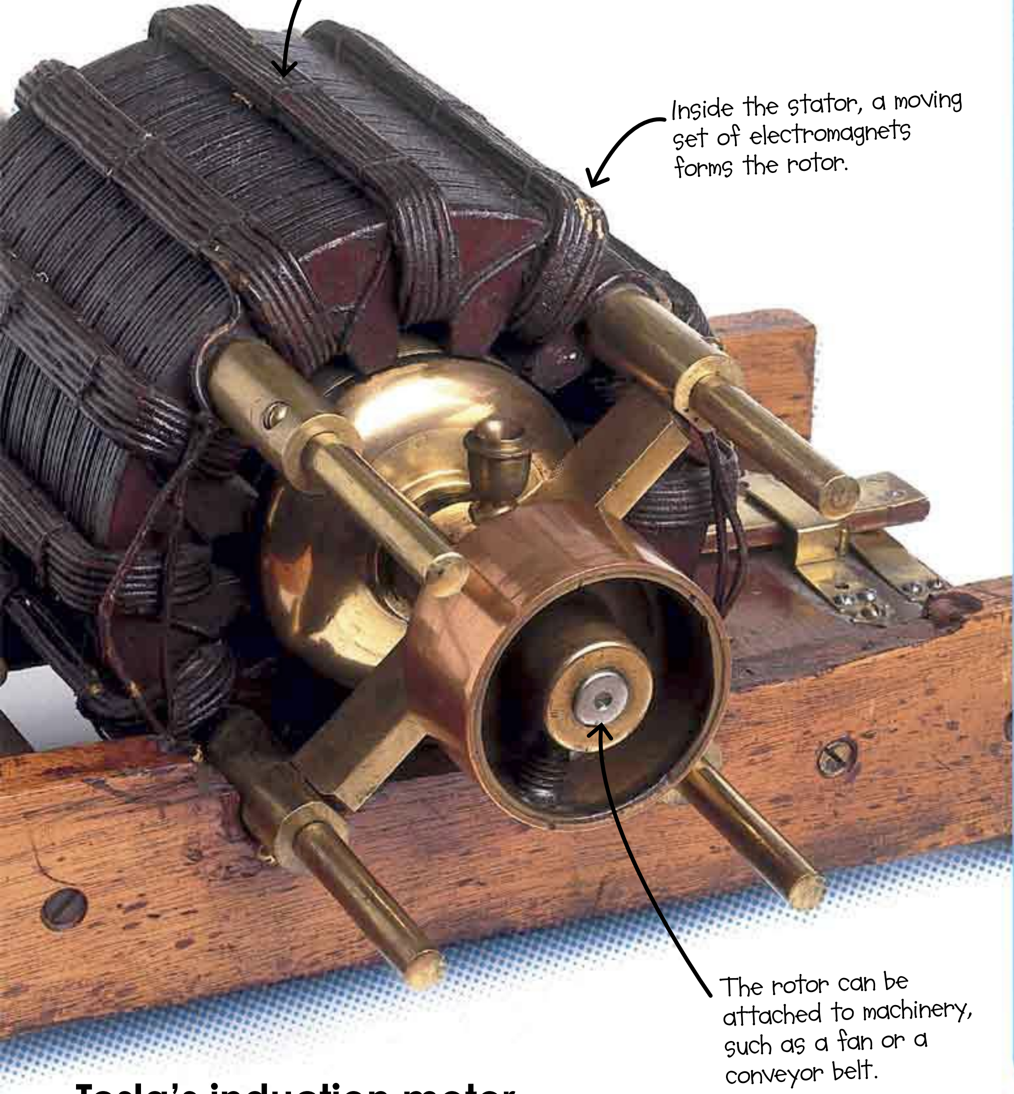
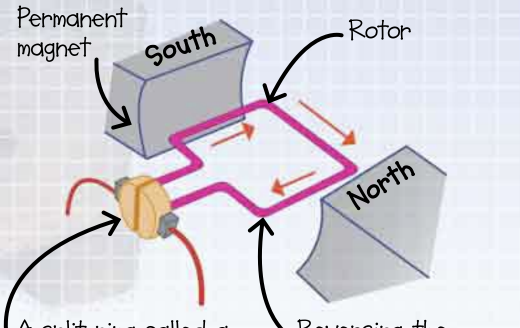


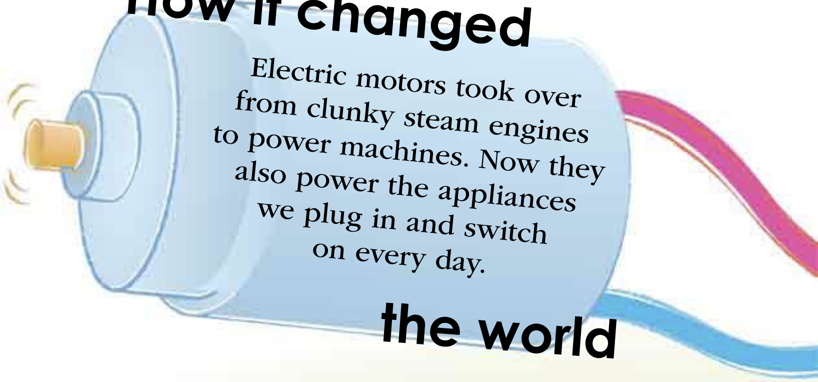
HOW IT WORKS

Motors that run on direct current have a permanent magnet and an electromagnetic rotor. The rotor's north and south poles are attracted to the opposite poles of the permanent magnet, so the rotor moves half a turn. The direction of the current is then reversed, so the rotor moves another half-turn. Continually switching the current like this keeps the motor spinning. Motors that use AC work in a similar way, but they do not need a mechanism to reverse the current.



Tesla's induction motor

Nikola Tesla invented the electric motors that power large machines today. His induction motor, invented in 1887, runs on alternating current (AC)—electric current that changes direction many times a second—rather than the direct current (DC) provided by a battery.



How it changed the world

Electric motors took over from clunky steam engines to power machines. Now they also power the appliances we plug in and switch on every day.

It paved the way for...

Steam-powered **WASHING MACHINES** *were laundering clothes in the 1800s, but electric motors made them smaller and more convenient.*

ELECTRIC CARS *were first invented in the 19th century, but only now do they look set to rival gas-powered ones.*